In fact the above discussion indicatesthat money supply becomes ***endogenous*** with fixed exchange rates and perfect capital mobility, in the sense that it passively adjusts to changes in the foreign exchange reserves as the central bank intervenes in foreign currency markets to defend the fixed parity. That is, since the central bank wants to hold the exchange rate constant, it has to intervene to keep the exchange rate from depreciating or appreciating, so foreign exchange reserves and hence money supply adjusts accordingly.

Our discussion also highlights the fact that it is impossible to simultaneously have the following three features in a single policy regime, viz.: (a) perfect capital mobility; (b) independent monetary policy; and (c) fixed exchange rates. This is also known as the '*policy trilemma*'. In the presence of perfect capital mobilityand *fixed* exchange rates, money supply becomes endogenous and the Central Bank's attempt to pursue an independent monetary policy isnot effectiveasthe Central Bank is committed to maintaining a fixed parity. Whereas, in the case of *flexible* exchange rates, perfect capital mobility and independent monetary policy do indeed go together, as we saw that monetary policy is very effective in this case. Countries may adopt fixed exchange rates and yet retain monetary policy independence only with restrictions on international capital flows, i.e., in the absence of perfect capital mobility. In this case, interest rate differentials between countries would persist owing to restrictions on capital mobility.

Check Your Progress 2

1. What is the effect of fiscal contraction (cut in tax rate) on output and interest rates in an open economy with perfect capital mobility under fixed exchange rate?

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2. Comment on the effectiveness of fiscal policy in open economies in light of your answer toquestion 1 above.

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3. (a) What is the effect of monetary contraction (sale of bonds by central bank) on output and interest rate in an open economy with perfect capital mobility under (i) fixed exchange rates, and (ii) flexible exchange rates?

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(b) Explain the concept of ‘endogeneity’ of money supply.

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(c) Comment on the effectiveness of monetary policy in open economies in light of your answer in part (a) above.

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4. Use the open economy IS-LM model to explain the concept of ‘policy trilemma’.

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5.5 LET US SUM UP

In this Unit we discussed the open economy IS-LM model wherein we saw that international trade affects aggregate demand in an economy, while capital flows play an important role in shaping policy outcomes through their impact on interest rates, exchange rates and money supply. In particular we saw that with fixed exchange rates and perfect capital mobility, money supply becomes endogenous, i.e., countries cannot conduct independent monetary policy. Fiscal policy is an effective policy tool when the exchange rate is fixed; however, with flexible exchange rates and perfect capital mobility, fiscal expansion becomes ineffective as it has no effect on output and leads to a worsening of the trade balance. Finally, the effectiveness of monetary policy under flexible exchange rates demonstrates that countries can have at most any *two* of the three policy choices of having *independent monetary policy, fixed exchange rates* and *perfect capital mobility*.

5.6 ANSWER/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) Refer to Sub-section 5.3.1 and Fig. 5.3.
2) Refer to Sub-section 5.3.1.
3) Refer to Sub-sections 5.2.2 and 5.2.3 and Fig. 5.1

Check Your Progress 2

1) Refer to Sub-section 5.4.1 and Fig. 5.4.
2) Refer to Sub-section 5.4.1 and Fig. 5.4.
3) a) Refer to Sub-section 5.3.2 and 5.4.2, and Fig. 5.3 and Fig. 5.5.
   b) Refer to Sub-section 5.4.2, and Fig. 5.5.
   c) Refer to Sub-sections 5.3.2 and 5.4.2, and Fig. 5.3 and Fig. 5.5.
4) Refer to Sub-section 5.4.2.